SAIKUMAR KALERU

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PROFESSIONAL SUMMARY

MBA in Finance and BTech in Computer Science, combining quantitative finance with expertise in low-latency systems and trading infrastructure. Designed and built ultra-low latency trading infrastructure for High-Frequency Trading targeting NSE equities. Strong expertise in derivatives and financial risk.

SKILLS

Programming: C, C++, Java, Python, JavaScript, TypeScript, SQL, DSA

AI & ML: Scikit-learn, TensorFlow, Deep Learning, Neural Networks, GEN AI, LLMs, RAG, LangChain, LangGraph, CrewAI Workflow Orchestration (n8n), MCP Server, Agentic AI, MLOps, Prompt Engineering,

Web Development: React, Next.js, Node.js, REST APIs, FastAPI

Systems & HFT: Linux, Low-Latency Architecture, Lock-Free Programming, Kernel Bypass, SPSC Queues, Memory Pools

Infrastructure: AWS, Git, Docker, CI/CD, Computer Networks

PROFESSIONAL EXPERIENCE

Member Technical *May 2022 – Jan 2024*

D. E. Shaw India Private Limited · Full-time

- Designed and implemented sub-microsecond low-latency systems for High-Frequency Trading (HFT) in C++.
- Implemented the NSE Non-NEAT Frontend (NNF) protocol for order entry, execution reports, and drop copy communication.
- Built market-making system: L3 order book, Avellaneda-Stoichkov pricing model, pre-trade risks (rate limiting, position, P&L collars), OMS/RMS with SPSC shared memory queues and kernel bypass via OpenOnload.
- Developed Order Management System (OMS) and Risk Management System (RMS) with confirmed position ledger, rejection monitor and EOD reconciliation.
- Ran historical tick-by-tick market data simulations; back tested and validated market-making strategies.
- Performed C++ performance optimization and analysis; Profiled hot-path execution to eliminate latency spikes.
- Built system infrastructure components including a lock-free logger for real-time event capture, latency measurements using histograms, and tick-by-tick data logging to record full system activity.
- Maintained and expanded the universe of stocks by onboarding new instruments; Managed infrastructure config changes including tick size, currency, and stock-level parameters.

Site Reliability Engineer *Jan 2022 – May 2022*

Arcesium India Private Limited · Internship

- Debugged and resolved system issues ensuring high availability of EC2 and RDS services.
- Managed MS SQL and PostgreSQL operations across uat, prod and test servers;
- Improved query optimization and schema management using the Flyway tool.

PROJECTS

Cache-efficient limit order book (C++)

- Designed a cache-efficient limit order book with O(1) order matching optimized for low-latency trading systems.
- Implemented memory-efficient data structures and custom allocation strategies to minimize heap overhead.
- Developed Python bindings for strategy prototyping and backtesting
- Built real-time visualization dashboard using WebSockets to display order book depth and trades

Designing a Resilient and Fault Tolerant Microservice Architecture for E-Commerce Application

- Designed a microservices-based architecture with independently deployable services for order management, payments, and user services

- Implemented containerized services using Docker and Kubernetes with service discovery and load balancing.
- Built CI/CD pipeline using Jenkins for automated build, testing, and deployment.

Implied Volatility Surface Modeling

- Built Python script to compute, visualize, and forecast implied volatility surfaces for option chain data.
- Retrieved live option chains via yfinance; Computed IVs using Black-Scholes & SciPy optimizers; Visualized 3D smiles/skews with Matplotlib.

EDUCATION

Master of Business Administration (MBA) – Finance *2024 – 2026*

University College of Commerce and Business Management, Hyderabad | CGPA: 8.9

B.E. in Computer Science and Engineering *2018 – 2022*

Chaitanya Bharathi Institute of Technology, Hyderabad | CGPA: 8.8

ACHIEVEMENTS

- Scored 95.6 percentile in UGC NET 2025 (Management) and top 5% in GATE CSE 2021.
- Ranked 212 in CodeChef long challenge coding contest (24,000+ participants).